

DEDEEPIYA AVANCHA

International Institute of Information Technology, Bangalore

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Education

International Institute of Information Technology Bangalore

Aug 2023 – Expected Jul 2028

BTech + MTech (Dual Degree) in Computer Science and Engineering

CGPA: 3.75/4.00 (9.3/10.0)

Experience

Tectoro Consulting Pvt. Ltd.

May 2025 – Jul 2025

Machine Learning Intern

Hyderabad, India

- Built an evaluation pipeline to compare real-time object detection models (**RT-DETR**, **YOLOv8**, **YOLOv9**) using **Python** and **Ultralytics**, selecting the optimal architecture for defect detection.
- Improved drone-based object detection accuracy from **81% to 92%** using **YOLOv9** with image slicing and optimized inference.

Projects

Kafka Benchmarking & Backpressure Control  | Go, Apache Kafka

Mar 2026 – Jun 2026

- Designed an adaptive **backpressure controller** with separate pause and resume lag thresholds to prevent oscillation, reducing peak consumer lag **96%** (309k to 12k messages) and end-to-end p99 latency **88%** (31.4s to 3.8s) under slow-consumer overload while maintaining **100% message delivery**.
- Built a modular Go Kafka benchmarking framework with a **lock-free producer path** (per-goroutine AsyncProducers, cache-line-padded atomics) sustaining **9.9k msg/s**; captured both broker-ack and end-to-end latency via 16-byte timestamped message headers across **10 decoupled packages**.
- Benchmarked across compression codecs, payload types, and burst/recovery workloads; found that **zstd** induces **16× higher GC cycle pressure** than alternatives.

AlgoLens  | Node.js, Go, gRPC, PostgreSQL

Apr 2026 – May 2026

- Built a search engine over **1,185 DSA problems**, implementing **BM25** and **TF-IDF** from scratch; BM25 outperformed TF-IDF, improving **MRR by 13.5%** and **nDCG@10 by 9.6%**.
- Implemented BM25 in both Node.js and a **Go gRPC microservice** behind a shared interface, verified **bit-exact ranking parity** and modeled user data in **PostgreSQL**.

Lock-Free SPSC Ring Buffer  | C++

Jan 2026 - Mar 2026

- Built a **single-producer single-consumer ring buffer** in C++ where two threads synchronize through only two **atomic** head/tail indices with **acquire-release memory ordering**, no locks, syscalls, or allocations on the hot path.
- Sustained about **380M ops/sec (2.6 ns per hand-off)**, roughly **13 to 15× faster** than a mutex-based queue; measured a **3× slowdown** from false sharing and a **2× speedup** from caching the reader/writer indices.
- Verified correctness with a race detector and a multi-million-message stress test, confirming zero data races and lossless hand-off.

Achievements

- Third Place, LinkedIn CoachIn Hackathon (2026):** Built CanvasAI, AI-powered learning software with an interactive canvas, knowledge graph, and active recall.
- Dean's Merit List Scholarship (2024–25):** Rs. 50,000 tuition fee waiver for academic excellence.
- Winner, Synergy:** IIIT Bangalore's flagship techfest hackathon (Neural Net track), 2025.
- Third Place, Hacknite Hackathon:** IIIT Bangalore (Apr 2024).

Technical Skills

Languages: Python, C++, Go, JavaScript, SQL, Java

Technologies/Frameworks: Node.js, PostgreSQL, PyTorch, LangChain, LangGraph, React, CMake, Git

Core Concepts: Data Structures & Algorithms, Machine Learning, Operating Systems, Database Management

Leadership & Extracurriculars

Teaching Assistant | IIIT-B

Aug 2025 – Dec 2025

- Mentored students in Linear Algebra through weekly tutorial sessions and graded exams.

E-Cell Ideation Team | Core Member

Sep 2024 – May 2025

- Designed and conducted workshops on finance for the student community.

Relevant Coursework

Data Structures & Algorithms, Design & Analysis of Algorithms, Operating Systems, Database Systems, Computer Networks, Machine Learning, Introduction to Optimization, Network-based Computing for HPC & AI, Software Engineering, Automata Theory & Computability, Computer Architecture, Discrete Mathematics